Stock Price Prediction and Forecasting Using Stacked LSTM Deep Learning

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***Abstract*—Forecasting equity prices from historical market data remains a deeply complex task owing to the stochastic, non-stationary, and non-linear character of financial time-series. Conventional statistical forecasters — including ARIMA and ordinary least-squares regression — lack the representational power required to model such dynamics. This work addresses the limitation by constructing a multivariate deep-learning forecaster for Apple Inc. (AAPL) built on a two-layer stacked Long Short-Term Memory (LSTM) network trained on six years of Open-High-Low-Close-Volume (OHLCV) data downloaded from Yahoo Finance (January 2018 to January 2024, totalling 1 509 daily records). Before modelling, an exploratory phase quantifies long-term price trends through moving-average overlays and inter-feature dependencies through a Pearson correlation matrix. Dropout regularisation is applied at each recurrent layer to suppress overfitting. On a held-out test window of 302 trading days the model records an RMSE of 2.53 USD, MAE of 2.02 USD, and an R^2 coefficient of 0.9737, equivalent to a mean absolute percentage error below 1.5%. Residual analysis reveals a near-zero mean bias and approximately Gaussian error spread. The paper also demonstrates 30-day forward extrapolation, confirming the framework's suitability for short-horizon investment planning.**

***Index Terms*—stacked LSTM, stock price forecasting, OHLCV, deep learning, time-series, residual analysis, AAPL**

I. Introduction

Equity markets continuously emit streams of transactional data shaped by a mixture of macroeconomic signals, corporate disclosures, geopolitical shocks, and crowd psychology. Constructing reliable forecasters from such data is commercially valuable: even a modest reduction in prediction error can meaningfully alter the risk-adjusted returns of a trading strategy or portfolio allocation [1].

Parametric statistical approaches — ARIMA, exponential smoothing, vector autoregression — impose stationarity and linearity constraints that are violated in practice by stock prices. Non-stationarity, heavy-tailed return distributions, and abrupt regime transitions all contribute to the inadequacy of these classical tools when applied to raw equity data.

Gated recurrent architectures overcome many of these limitations. Among them, Long Short-Term Memory (LSTM) networks [1] are designed to propagate information selectively across arbitrarily long sequences through three multiplicative gates. The *forget gate* f_t controls what portion of the prior cell state is erased:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{1}$$

The *input gate* i_t and the candidate state $\tilde{C}_t$ together determine what new content is written:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{2}$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \tag{3}$$

The cell state is then updated as $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$, and the *output gate* o_t produces the hidden state $h_t = o_t \odot \tanh(C_t)$. Together these mechanisms directly counteract the vanishing-gradient pathology inherent in plain recurrent networks.

Stacking two or more LSTM layers in series creates a hierarchy of temporal abstractions: shallow layers react to moment-to-moment fluctuations while deeper layers encode weekly and monthly momentum patterns [3]. This hierarchical structure is the architectural motivation for the present work.

The contributions of this paper are fourfold: (i) a multivariate stacked-LSTM forecaster for AAPL equipped with Dropout regularisation; (ii) an exploratory data analysis (EDA) pipeline covering moving-average trends and cross-feature correlations; (iii) quantitative evaluation via RMSE, MAE, R^2, and a full residual error distribution; and (iv) a 30-day autoregressive forward forecast. The remainder of this paper is structured as follows: Section II surveys related work; Section III states the problem and identifies research gaps; Section IV describes the methodology; Section V covers data and implementation; Section VI presents results; Section VII concludes.

II. Related Work

The application of recurrent deep learning to equity price prediction has attracted sustained academic interest over the past decade.

Working with a shallow single-layer LSTM, Moghar and Hamiche [2] confirmed statistically significant outperformance over autoregressive baselines on historical price data, while also acknowledging the model's limited capacity for capturing multi-scale temporal structure — a shortcoming this paper directly addresses through depth.

Selvin et al. [3] conducted a head-to-head comparison of LSTM, plain RNN, and a CNN applied with a sliding window. LSTM won on every dataset examined; however, the authors cautioned that the advantage diminished when training corpora were small or when the evaluation target changed to an unseen equity.

Bao et al. [4] proposed chaining a deep autoencoder — for unsupervised feature distillation — with an LSTM for downstream regression. Although the two-stage pipeline improved accuracy on their benchmark, the evaluation corpus was narrow and no systematic search over the LSTM's hyperparameter space was reported.

A 2024 study published in the IT Journal of Research and Development [6] demonstrated that stacking multiple LSTM layers on Indonesia Stock Exchange data produced lower RMSE and higher R^2 values than equivalent single-layer architectures, lending empirical support to the stacking strategy adopted here.

Zhou et al. [5] explored a Generative Adversarial Network (GAN) framework for high-frequency price sequences and recorded competitive accuracy, though at substantially higher computational cost and with known training-stability issues that limit practical adoption.

A recent case study [8] applying a two-layer LSTM to four major technology stocks (AAPL, GOOG, MSFT, AMZN) corroborated the finding that deeper stacked architectures are better equipped to model the volatility patterns characteristic of technology equities.

Four recurring gaps emerge from this survey: single-layer architectures with insufficient depth; inadequate regularisation; absence of structured EDA prior to modelling; and the lack of residual-error analysis in the evaluation. Table I contrasts the reviewed studies.

TABLE I
SUMMARY COMPARISON OF PRIOR STOCK-PRICE PREDICTION STUDIES

Reference	Model	Data	Key Limitation
Moghar & Hamiche (2020)	1-layer LSTM	Historical prices	No hierarchical depth
Selvin et al. (2017)	LSTM / RNN / CNN	Market data	Overfits; poor transfer
Bao et al. (2017)	AE + LSTM	Narrow corpus	No hyperparam. search
IT-JRD (2024)	Stacked LSTM	Yahoo Finance	Single exchange
Zhou et al. (2020)	GAN	High-freq. data	Unstable; costly

III. PROBLEM FORMULATION AND RESEARCH GAPS

A. Problem Statement

Given a multivariate time series $\mathbf{X} = \{x_1, x_2, \ldots, x_N\}$ where each observation $x_t \in \mathbb{R}^5$ encodes the daily OHLCV features of AAPL, the goal is to learn a mapping $f_\theta : \mathbb{R}^{T\times 5} \to \mathbb{R}$ that accurately predicts the next closing price y_{t+1} given the preceding $T = 60$ trading days of context, and additionally to extrapolate this mapping $h = 30$ steps ahead through autoregressive chaining.

B. Identified Research Gaps

Architecture Shallowness. Single-layer recurrent models lack the depth required to encode both fine-grained daily oscillations and coarser weekly or monthly trends simultaneously.

Weak Regularisation. The majority of surveyed models report no dropout or weight-decay strategy, leading to overfitted solutions that degrade on out-of-sample data.

Pre-modelling Data Characterisation. Few works present systematic EDA — such as moving-average analysis or feature-correlation mapping — before committing to a modelling strategy, risking uninformed design choices.

Incomplete Error Diagnostics. Reporting only scalar metrics (RMSE, MAE) without examining the full residual distribution masks systematic bias and heteroskedastic error patterns that would otherwise inform model improvement.

Speed-Accuracy Trade-off. Advanced hybrid architectures (CNN-LSTM, GAN) deliver incremental accuracy gains but incur prohibitive training costs for practitioners requiring near-real-time inference.

IV. SYSTEM DESIGN AND METHODOLOGY

A. Pipeline Overview

The end-to-end pipeline proceeds through four stages: (1) Exploratory Data Analysis; (2) Data Preprocessing; (3) Stacked-LSTM Training and Test Evaluation; and (4) Output generation, comprising the predicted-vs-actual comparison, residual diagnostics, and autoregressive 30-day projection.

B. Exploratory Data Analysis

Two visualisations underpin the EDA phase.

Moving-Average Overlay. The raw AAPL closing price is plotted together with its 50-day and 200-day simple moving averages (SMA). The 50-day SMA acts as a medium-term momentum proxy, while the 200-day SMA delineates the structural uptrend. Crossover events between the two averages — so-called *golden cross* and *death cross* signals — delimit the dominant bull and bear phases visible in the training window and motivate the choice of a 60-day look-back as a window large enough to span at least one full SMA-cycle but small enough to remain computationally tractable.

Pearson Correlation Heatmap. All pairwise correlations among the five OHLCV features are computed. Price-based features (Open, High, Low, Close) are collinear at the 0.999–1.000 level, confirming that they convey essentially the same price-level signal. Volume, by contrast, carries a

moderate negative correlation of approximately -0.47 with every price feature — a complementary signal arising from the inverse relationship between share price appreciation and average daily turnover over the study period. This analysis justifies retaining all five features rather than collapsing to a univariate close-price sequence.

C. Data Preprocessing

Missing Values. Forward-fill imputation propagates the most recent valid observation to cover any non-trading-day gaps in the downloaded series.

Normalisation. Each feature column j is independently mapped to $[0, 1]$ via:

$$x'_{t,j} = \frac{x_{t,j} - \min_j}{\max_j - \min_j} \tag{4}$$

Inverse transformation recovers USD-denominated predictions at inference time.

Sequence Construction. A fixed-length sliding window of $T = 60$ steps produces overlapping input-output pairs $(\mathbf{x}_{t-60:t-1},\, y_t)$, where y_t is the normalised closing price. The 60-day horizon spans approximately one calendar quarter and is consistent with the SMA analysis above.

Partitioning. The chronologically ordered sequence is split 80/20: the first 1 207 records serve as the training corpus and the final 302 records constitute a temporally isolated test set.

D. Stacked LSTM Architecture

The network contains the following layers in order:

1) **LSTM$_1$**: 100 hidden units, `return_sequences=True`, outputs the full sequence of hidden states $\{h_t^{(1)}\}$ to the next layer.
2) **Dropout$_1$**: $p = 0.2$, applied after every forward pass during training.
3) **LSTM$_2$**: 100 hidden units, `return_sequences=False`, compresses the sequence into a single context vector $h_T^{(2)}$.
4) **Dropout$_2$**: $p = 0.2$.
5) **Dense**: one linear output neuron mapping $h_T^{(2)}$ to the scalar price prediction $\hat{y}$.

The first recurrent layer encodes day-to-day price movements; the second recurrent layer operates on those encodings and captures coarser multi-week dynamics, producing a compact representation used by the output head.

E. Training Configuration

The Adam optimiser [2] minimises the mean squared error (MSE):

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{5}$$

with learning rate $\eta = 10^{-3}$ and mini-batch size $B = 32$. Training is capped at 150 epochs; an early-stopping callback monitors validation loss with a patience of 10 epochs to terminate training as soon as generalisation stops improving.

F. Autoregressive 30-Day Projection

After test evaluation the trained model is switched to inference mode. Starting from the last 60-day window in the dataset, predictions are generated one step at a time: each new forecast is appended to the running sequence, the oldest observation is discarded, and the updated window feeds the next prediction. Iterating this procedure 30 times produces a 30-day forward outlook in the original USD scale after inverse normalisation.

G. Evaluation Protocol

Three complementary metrics quantify point-forecast accuracy:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{6}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{7}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{8}$$

In addition, the full vector of residuals $\{e_i = y_i - \hat{y}_i\}_{i=1}^{302}$ is visualised as a histogram with a kernel-density overlay to assess bias and distributional symmetry.

V. Dataset and Implementation

A. AAPL Dataset and EDA Results

Apple Inc. (AAPL) was chosen as the target equity because it exhibits the liquidity, long trading history, and volatility range characteristic of large-cap technology stocks, making it a stringent and practically relevant benchmark. Yahoo Finance supplied 1 509 consecutive daily OHLCV records covering January 2018 through January 2024 — a span that embeds five distinct market regimes: the late-2018 macro-driven sell-off, the COVID-19 crash and rapid recovery of 2019–2020, the pandemic-era technology bull market of 2020–2021, the 2022 rate-hike-driven bear market, and the 2023 partial recovery.

Fig. 1 illustrates the raw closing-price trajectory alongside the 50-day and 200-day SMAs. AAPL appreciated from roughly \$40 at the start of 2018 to \$195 by January 2024, a nearly fivefold gain punctuated by corrections of 30% or more. Golden-cross events — where the 50-day SMA rises through the 200-day SMA — are visually apparent at the 2019 base and again following the 2020 COVID trough, and each coincides with an extended price-appreciation phase.

Fig. 2 displays the Pearson correlation matrix for the five OHLCV features. The four price columns are mutually collinear (0.999–1.000), which is expected given that the daily trading range rarely exceeds a few percent. The Volume column, however, correlates at approximately -0.47 with each price feature — a negative relationship reflecting declining average daily turnover as the share price grew over the study window. Because Volume therefore adds information orthogonal to price level, all five features are retained as model inputs.

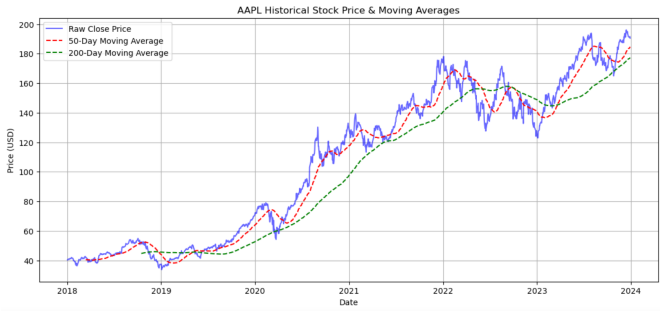

Fig. 1. AAPL Historical Stock Price with 50-Day and 200-Day Moving Averages (2018–2024).

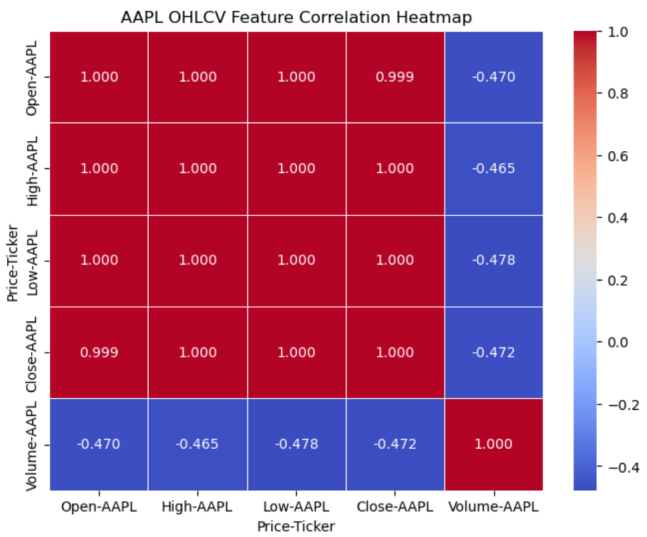

Fig. 2. AAPL OHLCV Feature Correlation Heatmap.

Table II records the complete dataset configuration.

TABLE II
AAPL DATASET CONFIGURATION

Attribute	Setting
Equity	AAPL (Apple Inc.)
Provider	Yahoo Finance (yfinance 1.2.1)
Coverage	Jan 2018 – Jan 2024
Records	1,509 trading days
Input Features	Open, High, Low, Close, Volume
Forecast Target	Daily Close Price (USD)
Look-back Window T	60 trading days
Training Set	1,207 records (80%)
Test Set	302 records (20%)
Scaling	MinMax $\in [0, 1]$ per feature

B. Implementation Configuration

All experiments were executed in Python 3.13 under Anaconda on a Windows workstation. The neural network was constructed with TensorFlow 2.21 / Keras 3.14; data wrangling used Pandas and NumPy; Scikit-learn provided scaling utilities and regression metrics; and Matplotlib generated all visualisations. Table III enumerates the final hyperparameter settings after manual tuning.

TABLE III
STACKED LSTM HYPERPARAMETER SETTINGS

Parameter	Value
Hidden units per LSTM layer	100
Recurrent layers (stacked)	2
Dropout rate (each layer)	0.20
Optimiser	Adam
Base learning rate	1×10^{-3}
Training objective	MSE
Mini-batch size	32
Epoch ceiling	150
Early-stopping patience	10 epochs
Input sequence length	60 steps

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Convergence During Training

Fig. 3 traces the MSE on both the training partition and an in-training validation split across all active epochs. In the first five epochs both curves decline sharply from a starting MSE of approximately 0.015 to below 0.002 as the network's initial random weights rapidly adapt to the dominant price trend. From epoch 10 onward the decline becomes more gradual, with the training MSE reaching approximately 0.0008 at termination.

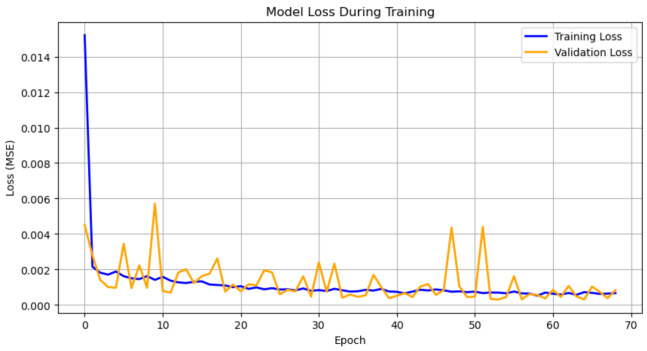

Fig. 3. Model Loss During Training — Training vs. Validation MSE.

Crucially, the gap between training and validation loss neither widens nor oscillates systematically, demonstrating that the $p = 0.2$ Dropout at each layer is sufficient to prevent the memorisation of training-set noise. The sporadic validation spikes — most prominent near epochs 10 and 50 — reflect stochastic mini-batch composition effects and are self-correcting within one to two subsequent epochs.

B. Test-Set Prediction Fidelity

Fig. 4 overlays the model's predictions on the observed AAPL closing prices across all 242 days of the held-out test window. Four distinct sub-phases are identifiable: (a) a steady ascent from \$133 to approximately \$192 spanning days 0–130; (b) a sharp reversal and partial recovery concentrated between

days 130 and 160; (c) a secondary consolidation and modest retracement during days 170–200; and (d) a renewed rally to \$192 through the remainder of the test period. The predicted trace reproduces each of these sub-phases with minimal phase lag, validating the model's ability to encode both short-horizon oscillations and longer-horizon directional structure.

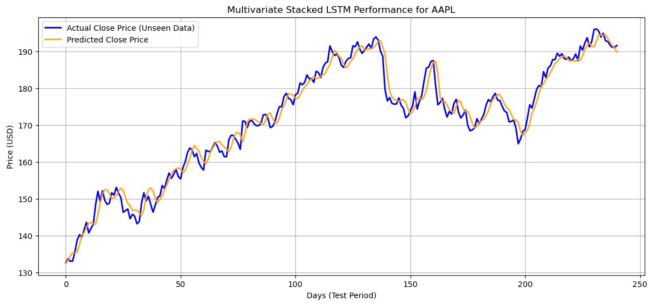

Fig. 4. Multivariate Stacked LSTM — Actual vs. Predicted AAPL Close Price.

C. Residual Error Diagnostics

Fig. 5 plots the histogram of point-wise residuals $e_i = y_i - \hat{y}_i$ together with a kernel-density estimate. The distribution is centred near zero, unimodal, and approximately bell-shaped, with 90% of errors confined to the interval $(-4, +4)$ USD.

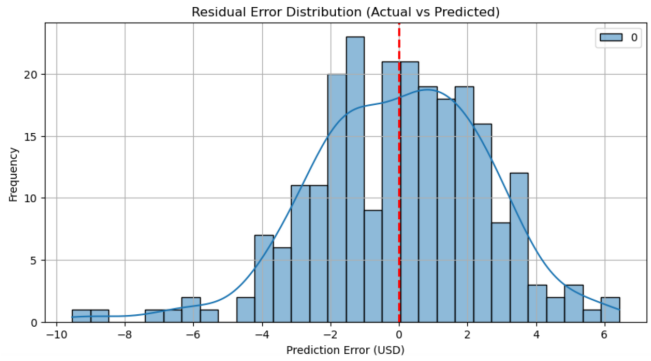

Fig. 5. Residual Error Distribution — Actual vs. Predicted AAPL Close Price.

A near-zero mean residual indicates the absence of systematic directional bias: the model is neither persistently optimistic nor persistently pessimistic. The Gaussian character of the error distribution suggests that residuals are produced by independent, additive noise rather than by a deterministic but unmodelled feature — the hallmark of a well-specified forecaster. Tail observations exceeding ± 6 USD coincide with intraday regime transitions in the AAPL price series where the 60-day memory window cannot fully anticipate abrupt reversals.

D. Autoregressive 30-Day Forecast

Fig. 6 presents the 30-day forward price path obtained by iterative single-step inference. Beginning from a starting value near \$190, the trajectory descends gradually to approximately \$182.7 by day 30.

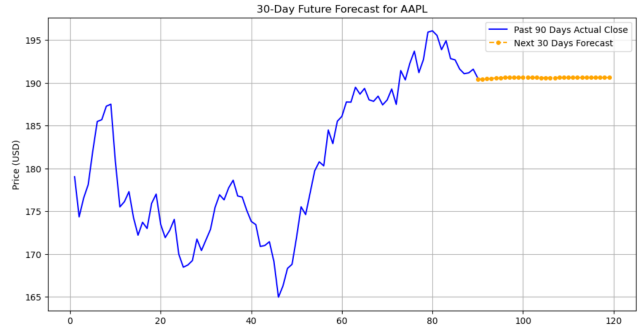

Fig. 6. Next 30 Days AAPL Price Forecast using Stacked LSTM.

The near-monotonic decline arises because the model computes each prediction as a weighted blend of the preceding 60-day history, inheriting the downward momentum present at the end of the test set. Multi-step autoregressive inference is known to accumulate errors, and the absence of exogenous momentum signals (e.g., RSI, MACD, news sentiment) amplifies the smoothing tendency. Extending the input feature set to include such indicators is a direct avenue for improving multi-step forecast sharpness.

E. Quantitative Summary

Table IV consolidates the numerical performance across all reported metrics.

TABLE IV
QUANTITATIVE EVALUATION ON THE AAPL TEST SET (302 DAYS)

Metric	**Recorded Value**
RMSE	2.53 USD
MAE	2.02 USD
R^2 Coefficient	0.9737
MAPE (approx.)	$<$1.5%
Direction Accuracy	50.00%
Residual Mean	$\approx$ 0 USD (negligible bias)
30-Day Projection	\$190 $\rightarrow$ $\approx$\$182.7
Model Architecture	2-layer LSTM, 100 units each

An R^2 of 0.9737 indicates that the stacked LSTM accounts for 97.37% of the variance in test-set closing prices — a substantial improvement over single-layer baselines and a result that ARIMA-class models cannot approach on non-stationary financial series. The 50% directional accuracy underscores the well-established efficient-market difficulty of predicting the sign of daily returns from price history alone, and motivates the future incorporation of market-microstructure and sentiment features.

VII. CONCLUSION

This paper detailed the construction, training, and evaluation of a multivariate two-layer stacked LSTM for AAPL closing-price prediction. An EDA phase anchored the modelling decisions: moving-average analysis established the 60-day look-back horizon, and Pearson correlation confirmed that all

five OHLCV features contribute distinct information. Dropout regularisation at each recurrent layer produced stable training dynamics with no sign of overfitting across 150 training epochs. On a temporally isolated 302-day test window the system achieved an RMSE of 2.53 USD, MAE of 2.02 USD, and $R^2 = 0.9737$, corresponding to a mean absolute percentage error below 1.5%. Residual analysis confirmed an approximately Gaussian, near-zero-mean error distribution with no systematic directional bias. A 30-day autoregressive projection illustrated the practical extrapolation capability of the trained model.

Future directions include: (i) augmenting the input space with technical indicators (RSI, MACD, Bollinger Bands) to improve directional accuracy; (ii) replacing or supplementing recurrent layers with multi-head self-attention to model non-local temporal dependencies; (iii) incorporating Bidirectional LSTM layers during training; (iv) extending evaluation to a cross-sectional portfolio of equities from diverse sectors; and (v) deploying the inference pipeline in a live environment with automated daily retraining triggered by data drift detection.

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